

Crowd and Preference Aware Indoor Path Planning

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Abstract—Indoor navigation in large public facilities requires path planning that accounts for accessibility constraints and crowd conditions, beyond shortest-distance path planning. We study crowd- and preference-aware indoor path planning on LiDAR-derived floorplans with static obstacles, where infeasible regions are masked based on user accessibility requirements. Crowd effects are modeled using synthetic density fields with controllable heterogeneity, including homogeneous and clustered crowd distributions. We benchmark Dijkstra, A*, and the Fast Marching Method (FMM) under two objectives: distance-only path planning and crowd-aware path planning using a density-penalized cost formulation. Using a simulator, we evaluate path quality and efficiency across grid resolutions and crowd heterogeneity levels. Results show that crowd-aware planning consistently reduces exposure to high-density regions with modest increases in path length and travel time. In particular, crowd-aware FMM achieves the strongest reduction in mean path density (up to 50.52%), compared to crowd-aware Dijkstra (up to 24.21%) and crowd-aware A* (up to 20.87%). These findings quantify key trade-offs between efficiency and crowd avoidance and provide guidance for selecting indoor path planning methods under accessibility and crowd considerations.

Index terms— Indoor navigation, path planning, crowd-aware path planning, accessibility-aware navigation.

I. INTRODUCTION

Indoor navigation is increasingly required in large and complex public facilities such as airports, hospitals, shopping malls, and transit stations. In these environments, classical shortest-path navigation is often insufficient because path quality depends on factors beyond geometric distance, including static obstacles, physical accessibility constraints, and crowding conditions. When crowding and accessibility are not considered, distance becomes the dominant criterion for path selection [1]. However, practical indoor navigation requires computing paths that remain feasible for different mobility profiles (e.g., wheelchair users) and that can reduce exposure to crowded areas for safety and comfort.

Crowd density is a key determinant of path quality in high-traffic indoor venues. In crowded environments, the shortest path is not necessarily the fastest or safest path. High-density areas may cause delays, discomfort, and in extreme cases lead to hazardous situations, as observed in mass gatherings [2]. Therefore, indoor path planning should support navigation that explicitly trades off path length with reduced exposure to crowded regions, while maintaining feasibility under accessibility constraints.

In this work, we study crowd- and preference-aware indoor path planning on Light Detection and Ranging (LiDAR)-derived indoor maps with static obstacles. Here, *preference-aware* refers to (i) *accessibility preferences* that impose feasibility constraints (e.g., stairs are infeasible for wheelchair users), and (ii) *objective preferences* that specify whether the user prioritizes shortest-distance path planning or crowd-aware path planning. Our pipeline constructs a navigable grid representation from LiDAR-based floorplan capture, applies accessibility-based masking to remove infeasible regions, and models crowds using synthetic density fields with controllable heterogeneity (homogeneous versus clustered distributions). Crowd-aware planning incorporates density as a penalty term in the path cost. We benchmark three widely used planning methods: Dijkstra [3], A* [4], and the Fast Marching Method (FMM) [5], using a simulator that enables controlled evaluation across grid resolutions and crowd patterns.

This study is guided by the following research questions:

- **RQ1:** How does incorporating crowd density into the planning cost affect the trade-off between path length and exposure to crowded regions?
- **RQ2:** How does crowd heterogeneity (homogeneous versus clustered distributions) influence path selection and the effectiveness of crowd-aware planning?
- **RQ3:** How do Dijkstra, A*, and FMM compare in computation time and path quality for indoor path planning?

This paper makes the following contributions:

- 1) We propose a crowd- and preference-aware indoor navigation pipeline that integrates LiDAR-derived maps, static obstacles, and accessibility constraints for feasible path computation.
- 2) We implement a controllable crowd generation model producing homogeneous and heterogeneous crowd distributions to support systematic evaluation.
- 3) We provide an experimental comparison across multiple grid resolutions and crowd heterogeneity levels, reporting path quality and computational performance.

Our evaluation highlights clear trade-offs between distance-only and crowd-aware objectives. Incorporating crowd density into the planning cost consistently reduces exposure to crowded regions, with algorithm-dependent effects on path length and travel time (**RQ1**). In particular, C-FMM

(‘C’ denotes crowd-aware) achieves the strongest crowd avoidance, reducing mean path density by up to 50.52% compared to baseline FMM, at the cost of longer detours (up to 20.60% increase in path length in our settings). In contrast, C-A* and C-Dijkstra primarily perform local avoidance, yielding moderate density reductions (up to 20.87% and 24.21%, respectively) with negligible change in geometric path length. We further observe that clustered crowd distributions amplify differences between distance-only and crowd-aware path planning (**RQ2**). Finally, under distance-only path planning, all three methods produce comparable shortest paths, while A* achieves the lowest computation time due to heuristic-guided search (**RQ3**).

II. RELATED WORK

Indoor navigation has received increasing attention due to its importance in large public facilities such as airports, hospitals, and shopping malls. Unlike outdoor navigation, indoor navigation cannot rely on GPS and must operate on building-specific representations that capture indoor connectivity, geometry, and constraints. Early work highlighted challenges such as limited signal availability and localization errors [6], [7].

Classical path planning approaches such as Dijkstra [3] and A* [4] compute optimal or near-optimal shortest paths on discrete graphs and grids. These methods are widely used in indoor settings due to their efficiency and flexibility in incorporating constraints via edge or cell costs. However, distance-only path planning does not capture key factors that affect practical indoor movement, including accessibility limitations and crowd.

To model navigation over continuous cost fields, the Fast Marching Method (FMM) has been used to compute geodesic shortest paths by solving the Eikonal equation [5]. Melchior et al. [8] compared A* and FMM for safety-aware navigation by incorporating obstacle danger levels using a potential-field formulation, showing that FMM can generate smoother paths that better approximate continuous optimal trajectories. Unlike our work, these studies do not incorporate crowd density as a planning factor.

Crowd-aware navigation extends indoor path planning by accounting for the effect of human density on travel time, comfort, and safety. Liu et al. [9] proposed a crowd-aware indoor path planning framework supporting queries such as Fastest Path and Least Crowded Path, emphasizing time-evolving population estimation and query processing over indoor networks. More recently, Liu et al. [10] investigated pedestrian path choice in crowded indoor environments using stepwise re-planning under uncertainty in crowd observations. While these studies motivate crowd-aware planning as a necessary extension to distance-only path planning, prior work often focuses on network-level query processing with crowd prediction [9] or dynamic re-planning and behavioral modeling [10]. In contrast, our work focuses on controlled benchmarking of crowd-aware cost formulations across multiple planning algorithms under different crowd heterogeneity settings.

Accessibility constraints are also essential for indoor navigation systems intended for diverse users, including wheelchair users and individuals with reduced mobility. Cardoso et al. [11] incorporated accessibility-related building features into a grid-based indoor navigation framework using A*-based planning with mobility-specific constraints. Similarly, our work integrates accessibility feasibility constraints, but evaluates how these constraints interact with crowd-aware objectives across different algorithms.

Indoor path planning performance depends strongly on the underlying map representation. Grid-based discretization is widely adopted because it enables both graph search and continuous propagation methods. Prior work has explored grid-based models for pedestrian tracking and indoor spatial processing [12]. Hexagonal grids have been proposed as an alternative discretization that improves neighborhood regularity and reduces directional bias compared to square grids [13]. In this work, we adopt a hexagonal grid representation derived from LiDAR-based floorplan capture to support systematic comparison across grid resolutions.

Overall, prior work provides strong foundations in shortest-path planning, continuous cost-field planning, crowd-aware navigation, and accessibility-aware path planning. However, there remains a gap in systematically benchmarking discrete planners (Dijkstra, A*) and continuous planners (FMM) under a unified crowd-aware cost formulation while simultaneously incorporating accessibility constraints. This paper addresses this gap through a simulator-driven evaluation framework over LiDAR-derived indoor maps and controlled crowd distributions.

III. METHODOLOGY

Given an indoor floorplan, the simulator constructs a discrete navigable representation in which each hexagonal grid cell stores attributes including traversability, accessibility, and crowd density. We support two mobility profiles: (i) non-disabled users (ND) and (ii) wheelchair users (WD). The profiles differ in (a) feasible connectors (e.g., stairs infeasible for WD) and (b) assumed walking speed used for travel-time estimation. We use the term *preference-aware* to refer to both *accessibility preferences*: feasibility constraints derived from the user mobility profile, and *objective preferences*: whether the user prioritizes distance-only path planning or crowd-aware path planning.

The simulator takes a LiDAR-derived floorplan as input and produces a navigable grid. Traversability is first determined by static obstacles (e.g., walls and furniture) and then modified by accessibility masking based on the user profile. Next, a crowd density generation module produces a static density field with controllable heterogeneity. Finally, given a start and destination, each algorithm computes the preferred path under either the distance-only objective or the crowd-aware objective.

A. Floor Mapping and Semantic Extraction

We acquire the indoor floorplan using the Polycam mobile LiDAR application [14] and export it in .dxf format. The

CAD file contains layered vector geometry with semantic annotations (e.g., walls, openings, elevators, stairs, furniture, fixtures), which enables differentiation between obstacles and traversable connectors.

Mobile LiDAR floorplans may include artifacts such as fragmented boundaries and small gaps due to occlusions and sensor noise [15]. We therefore apply preprocessing to (i) parse the CAD file, (ii) extract geometries per semantic layer, and (iii) clean and normalize the geometry (merge nearby polylines, repair gaps, and correct mislabeled layers). The output is a cleaned semantic floorplan suitable for discretization and traversability labeling.

B. Hexagonal Grid Tiling

Regular Hexagonal tiling provides isotropic connectivity: each cell has six equidistant neighbors, reducing directional bias compared to square grids [13]. We discretize the continuous floorplan into a regular hexagonal grid to support both graph-based planners (Dijkstra, A*) and continuous cost-field planning (FMM).

Given the floorplan boundary, we generate a staggered hexagonal lattice and place cell centers over the region of interest. As it is shown in Figure 1 the tessellation may not align perfectly with the map boundary, we extend the grid slightly beyond the floorplan extent and apply a small offset to avoid uncovered boundary regions. A higher grid resolution produces smaller cells, improving accuracy and better approximating human walking behavior, while increasing computational and time complexity.



Fig. 1: Partially covered area with 5×5 hexagonal grid

Each cell is assigned a base traversability cost using the semantic layers. Cells overlapping obstacles are blocked (cost ∞) and free-space/connectors are traversable (base cost 1). Accessibility constraints then modify this traversability mask per user profile (e.g., stair cells blocked for WD), ensuring that computed paths remain feasible.

C. User Modeling and Accessibility Constraints

User preferences are captured through feasibility constraints and evaluation parameters. We consider:

- **Non-disabled (ND)**: can traverse all traversable cells/connectors.
- **Wheelchair user (WD)**: cannot traverse stairs and must use accessible connectors (e.g., elevators).

a) *Accessibility masking.*: Let each cell c have base traversability $w(c) \in \{1, \infty\}$. For WD, stair cells are relabeled as blocked (∞), while ND retains the base mask.

b) *Walking speed.*: Each profile is associated with a representative walking speed used to estimate travel time from the computed path length (Section IV). This work assumes a static pedestrian crowd representation.

c) *Safety distance.*: We assume WD requires larger interpersonal spacing than ND. Since crowd is modeled via static density fields, the planner indirectly accounts for safety by avoiding high-density regions in crowd-aware mode.

D. Crowd Density Generation

To evaluate crowd-aware planning, we generate synthetic crowd distributions over the traversable space using a density-based clustering approach. [16]. Crowd is represented as sampled pedestrian locations that are aggregated to grid cells to produce a density value $p(c)$ (persons/m²).

We control crowd heterogeneity using a parameter $H \in [0, 1]$. When $H = 0$, pedestrians are distributed uniformly across traversable cells. Larger H produces increasingly clustered distributions with localized high-density regions. For each planning run, the resulting density field is static and used as a penalty in crowd-aware cost functions.

E. Path Planning Algorithms

We benchmark three representative planning methods: Dijkstra [3], A* [4], and FMM [5]. Each algorithm is evaluated under two objectives. We denote the crowd-aware variants as C-Dijkstra, C-A*, and C-FMM.

- **Distance-only path planning** (baseline): minimize geometric distance.
- **Crowd-aware path planning**: minimize a density-penalized cost that trades off distance and crowd exposure.

1) *Dijkstra Algorithm*: Dijkstra computes the shortest path from a start node to all other nodes in a weighted graph by expanding the frontier in increasing order of cumulative cost. Let $g(n)$ denote the cumulative distance cost from the start node to node n . The evaluation function is:

$$F(n) = g(n). \quad (1)$$

To incorporate crowd avoidance, we augment the traversal cost of each step using a density-dependent penalty. This is inspired by prior work that incorporates risk or danger levels into planning cost functions [8]. Let $d(n, n-1)$ denote the geometric distance between two adjacent grid cells and let p_n denote the crowd density of cell n (persons/m²). We define the crowd-aware step cost as:

$$c(n) = \sqrt{d(n, n-1)^2 + \lambda p_n^2}, \quad (2)$$

where λ controls the trade-off between path length and crowd avoidance. C-Dijkstra then minimizes the cumulative crowd-aware traversal cost:

$$F(n) = g(n) + c(n). \quad (3)$$

2) *A* Algorithm*: A* is a heuristic extension of Dijkstra that prioritizes exploration toward the goal using an admissible heuristic. Its evaluation function combines the cumulative path cost $g(n)$ with a heuristic estimate $h(n)$ of the remaining cost to the goal:

$$F(n) = g(n) + h(n). \quad (4)$$

Here, $g(n)$ denotes the exact distance cost accumulated from the start to n , and $h(n)$ is typically the Euclidean distance between n and the goal.

The crowd-aware variant C-A* uses the same crowd-aware step cost (Eq. 2) and augments the A* evaluation function accordingly:

$$F(n) = g(n) + c(n) + h(n). \quad (5)$$

This formulation encourages paths that remain short while reducing exposure to high-density regions.

3) *FMM Algorithm*: FMM computes shortest paths by solving the Eikonal equation for an arrival-time field $T(\mathbf{x})$:

$$|\nabla T(\mathbf{x})| G(\mathbf{x}) = 1, \quad (6)$$

$$|\nabla T(\mathbf{x})| = F(\mathbf{x}), \quad (7)$$

where $G(\mathbf{x})$ is the propagation speed. Under distance-only planning, $G(\mathbf{x})$ is constant over traversable regions and zero over obstacles. Where $F(\mathbf{x})$ is the propagation cost evaluated on the propagation front.

For crowd-aware planning, we define a density-dependent cost field that slows propagation in high-density regions:

$$c(\mathbf{x}) = 1 + \lambda p(\mathbf{x}). \quad (8)$$

$$|\nabla T(\mathbf{x})| = c(\mathbf{x}), \quad (9)$$

IV. EXPERIMENTS

A. Simulation Setup

We implemented the simulator in Python and evaluated all methods on a LiDAR-reconstructed indoor environment with a main and mezzanine level captured at the University of Calgary campus. Floorplans were acquired using an iPhone 16 Pro and the Polycam application [14] and exported in .dxf CAD format with semantic layers. The map includes static structural and semantic elements such as walls, furniture, fixtures, doors, elevators, and stairs, spanning an area of $50.56 \text{ m} \times 61.57 \text{ m}$. Table I summarizes the simulation parameters.

We consider two user mobility profiles: non-disabled (ND) and wheelchair user (WD). Accessibility constraints are applied through feasibility masking (e.g., stair regions are blocked for WD). Travel time is estimated using profile-specific walking speeds: $v_{\text{ND}} = 1.14 \text{ m/s}$ [17] and $v_{\text{WD}} = 0.5 \text{ m/s}$ [18]. These speeds are used only for evaluation and do not alter the path planning objective. Based on Canadian accessibility statistics [19], we assume a disability prevalence of 22%, yielding an average population speed of 0.9 m/s . Users are assumed to require interpersonal safety distances of 1.2 m (ND) [20] and 1.5 m (WD) [21]. Safety distance is not enforced as a collision constraint; instead, crowd effects are modeled via density penalties.

Crowding is modeled as a static density field over the grid. To generate crowd conditions, we sample and distribute 1,000 points across the environment where each point represents a pedestrian. The trade-off between distance and crowd

is controlled by a penalty weight λ . We show the λ s effect in the results section and use $\lambda = 30$ in our experiments.

Setting	Value
Area dimension	$50.56 \text{ m} \times 61.57 \text{ m}$
User types	Non-disabled, Wheelchair user
Speed (m/s)	$v_{\text{ND}} = 1.14, v_{\text{WD}} = 0.5$
Safety distance (m)	$d_{\text{ND}} = 1.2, d_{\text{WD}} = 1.5$
Penalty Weight (λ)	30
Grid shape	Regular hexagon
Grid availability	1, ∞
Crowd type & size	Static & 1,000
Area properties	Walls, Furniture, Fixtures, Doors, Elevator, Stairs
Access gate	Stairs, Elevators, Doors

TABLE I: Simulation setting

B. Experimental Factors

Table II summarizes experimental factors. We evaluate three grid resolutions (20×20 , 30×30 , 40×40) to study the impact of discretization granularity on path planning quality and runtime. Crowd heterogeneity is controlled using parameter $H \in \{0, 0.3, 0.5, 0.8\}$, where $H = 0$ produces a spatially uniform distribution and larger values yield increasingly clustered crowd regions. We cap H at 0.8 because the building layout provides limited alternative paths at higher clustering levels, leading to minimal additional variation in path choices.

For distance-only path planning, we run the baseline algorithms Dijkstra, A*, and FMM. For crowd-aware path planning, we run C-Dijkstra, C-A*, and C-FMM using the density-penalized objective described in Section III-E.

Factor	Value
Grid Resolution	$20 \times 20, 30 \times 30, 40 \times 40$
Heterogeneity Level (H)	0, 0.3, 0.5, 0.8
Algorithm	Dijkstra, A*, FMM
Objective	Distance, Crowd-Distance

TABLE II: Experiment factors

C. Experimental Procedure and Evaluation Metrics

All experiments are conducted on the same indoor environment captured from the University of Calgary main campus. While the map represents the same area, experiments are repeated with different start points, destinations, and crowd distributions.

To ensure fair comparison across planning methods, we evaluate all algorithms on identical start-destination pairs and crowd configurations. We generate crowd according to a heterogeneity level H , and then in each iteration, we randomly select one start cell and one destination cell from traversable regions. All six algorithms, A*, Dijkstra, FMM, C-A*, C-Dijkstra, C-FMM, are then executed using the same start point, destination point, grid resolution, and crowd realization. We repeat this process for $N = 30$ iterations with different start and destinations. In each iteration, we

record the identified path of each algorithm for comparison and analysis.

We quantify and compare path planning algorithms using the following metrics:

- *Path length (m)*: The total geometric length of the computed path from the start to the destination.
- *Travel time (s)*: The estimated traverse time of the path, computed using the path length and the assumed walking speed model described in Section III.
- *Computation time (s)*: The runtime required by each algorithm to compute a complete path.
- *Path density (persons/m²)*: We report the mean crowd density along the path, computed as the average density of all grid cells visited by the path.

To facilitate comparison between shortest-path and crowd-aware path planning, we also report relative changes with respect to the corresponding baseline algorithm (e.g., C-A* vs. A*). *Path Length Change (%)* describes the relative change in total path length. *Average Path Density Change (%)* measures relative change in mean density along the path. Finally, *Path Time Change (%)* reports relative change in traversal time.

V. RESULTS

A. General Observation: Map Loading and Penalty Effect

We first illustrate the simulator representation of the indoor environment and the generated crowd fields. Figure 2 shows the same LiDAR-derived map under the lowest and highest crowd heterogeneity levels used in our experiments. Traversable cells are color-coded by crowd density from low to high (green $\rightarrow$ yellow $\rightarrow$ orange $\rightarrow$ red). Blocked cells (e.g., walls, furniture, and fixtures) are shown in gray; stair regions are shown in darker gray and are additionally masked as infeasible for the wheelchair-user profile. Structural walls are outlined in blue. Cell centers are shown as red dots, and sampled pedestrian locations used to construct the density field are shown as black dots. As expected, $H = 0$ produces an approximately uniform distribution over traversable space, while $H = 0.8$ yields spatially clustered high-density regions that create localized crowded hot spots.

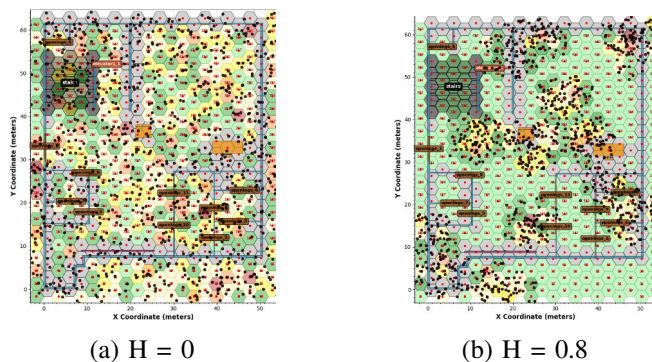


Fig. 2: Crowd distribution with different heterogeneity levels

Figure 3 illustrates the impact of the penalty weight λ on path selection (shown for a representative region of the

map). When $\lambda = 0$, the planner reduces to distance-only path planning and returns the geometric shortest path. As λ increases, crowd density contributes more strongly to the objective, and the planner progressively deviates from the shortest path to avoid high-density regions. Larger λ values therefore yield paths with lower exposure to crowded areas but longer geometric detours, highlighting the core trade-off between distance minimization and crowd avoidance.

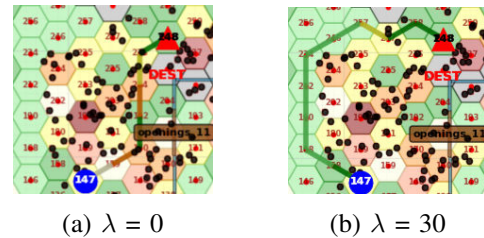


Fig. 3: Paths with and without crowd penalty

B. Algorithm Comparison with Shortest Distance Objective

We first compare Dijkstra, A*, and FMM under the *distance-only* objective by setting the crowd penalty weight to $\lambda = 0$. In this setting, crowd density does not influence path selection, and all algorithms aim to minimize geometric path length. To avoid introducing artificial crowd hotspots, we use a uniform crowd distribution on the map. Travel time is estimated using the population-average speed (0.9 m/s), derived from the assumed mobility-profile distribution.

Figure 5 reports path length, mean path density, and travel time for the baseline algorithms. As shown in Figure 5(a), all three methods produce paths with identical length across iterations, indicating that each finds a shortest path under the same discretization. While the selected paths are not always identical, density along the paths can differ slightly (Figure 5(b)) due to alternative shortest paths passing through different regions of the map. These density differences have a minor impact on travel time (Figure 5(c)).

Since path quality is comparable under distance-only path planning, computation time becomes a key differentiator. Figure 4 shows that runtime increases with grid resolution and map complexity for all methods. A* consistently achieves the lowest computation time because its heuristic-guided search reduces the number of explored nodes. In contrast, Dijkstra and FMM exhibit higher runtimes due to more exhaustive exploration/propagation.

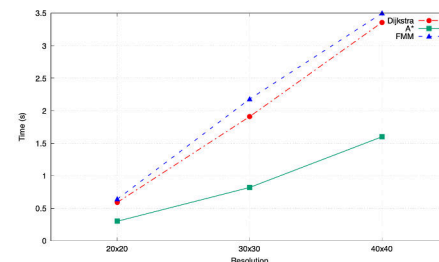


Fig. 4: Execution time of algorithms

Overall, when the user objective is shortest-distance path planning, A* provides the best practical choice: it returns shortest paths comparable to Dijkstra and FMM while offering significantly lower execution time.

C. Algorithm Comparison with Crowd & Distance Objective

We next compare algorithms under the *crowd-aware* objective, where path cost reflects both geometric distance and crowd density via a density-penalized formulation (Section III-E). We evaluate C-Dijkstra, C-A*, and C-FMM across crowd heterogeneity levels $H \in \{0, 0.3, 0.5, 0.8\}$ and compare each method against its distance-only baseline. Figure 6 summarizes *path length*, *mean path density*, and *travel time* for grid resolution 20×20 .

a) Impact on average path density: Across all heterogeneity levels, crowd-aware planning consistently reduces mean density along the computed path. C-FMM achieves the strongest reduction relative to FMM, decreasing mean path density by 34.06%, 48.57%, 45.71%, and 50.52% for $H = 0, 0.3, 0.5$, and 0.8 , respectively. Under the same conditions, C-Dijkstra reduces density by 12.08%, 19.04%, 24.47%, and 24.21% relative to Dijkstra, while C-A* reduces density by 13.97%, 19.12%, 18.91%, and 20.87% relative to A*. These results indicate that the crowd penalty is most effective in C-FMM, particularly under heterogeneous (clustered) crowd conditions.

b) Impact on path length: Density reduction can require deviating from the geometric shortest path, leading to longer routes. This effect is most pronounced for C-FMM, which increases path length by 5.88%, 20.60%, 13.73%, and 7.11% for $H = 0, 0.3, 0.5$, and 0.8 , respectively. The continuous cost-field formulation in FMM penalizes high-density regions throughout the domain, producing smooth global detours around crowded areas. In contrast, C-A* and C-Dijkstra show no statistically significant increase in path length in this setting. Their avoidance behavior is largely local, often selecting alternative paths with a similar number of traversed cells, and therefore nearly identical geometric length under uniform cell spacing.

c) Impact on travel time: Travel time follows both path length and crowd exposure. C-FMM increases travel time by 3.30%, 14.00%, 7.65%, and 2.59% relative to FMM across $H = 0, 0.3, 0.5$, and 0.8 , respectively, mainly due to longer detours. In contrast, C-A* and C-Dijkstra yield small travel-time reductions despite similar path lengths, because lower-density routes correspond to higher effective traversal speeds in our crowd model.

d) Execution Time: Crowd-aware variants introduce modest computational overhead compared to their baselines due to the density-augmented cost evaluation and expanded search for lower-density regions. Across grid resolutions, average slowdowns are approximately $0.13\times$, $0.18\times$, and $0.27\times$ for C-Dijkstra, C-A*, and C-FMM, respectively.

e) Comparing crowd-aware methods: Comparing the crowd-aware planners directly, C-FMM consistently produces the lowest-density paths, but at the cost of increased detours and travel time. User with the need to avoid crowd

should use C-FMM for their safety. Relative to C-A* and C-Dijkstra, C-FMM reduces average path density by 64.51% and 58.96% for $H = 0$, by 60.78% and 60.62% for $H = 0.3$, by 46.45% and 58.62% for $H = 0.5$, and by 52.08% and 58.67% for $H = 0.8$. This behavior reflects two distinct avoidance strategies: C-A* and C-Dijkstra reduce exposure primarily through local adjustments near the shortest path, whereas C-FMM exhibits more global avoidance by routing around dense regions.

Overall, the results reveal two distinct avoidance strategies. C-A* and C-Dijkstra primarily achieve crowd avoidance through *local* adjustments (choosing lower-density neighboring cells) while remaining close to the shortest path. In contrast, C-FMM exhibits a more *global* avoidance behavior by path planning around crowded regions, yielding larger density reductions but also higher path length and travel time in some scenarios.

VI. CONCLUSIONS AND FUTURE WORK

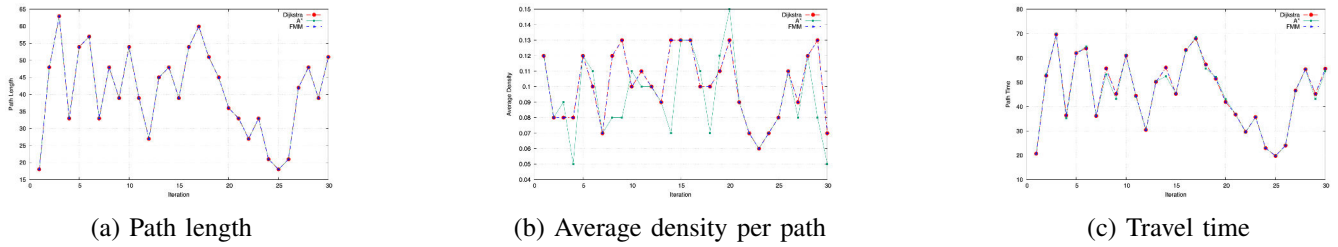
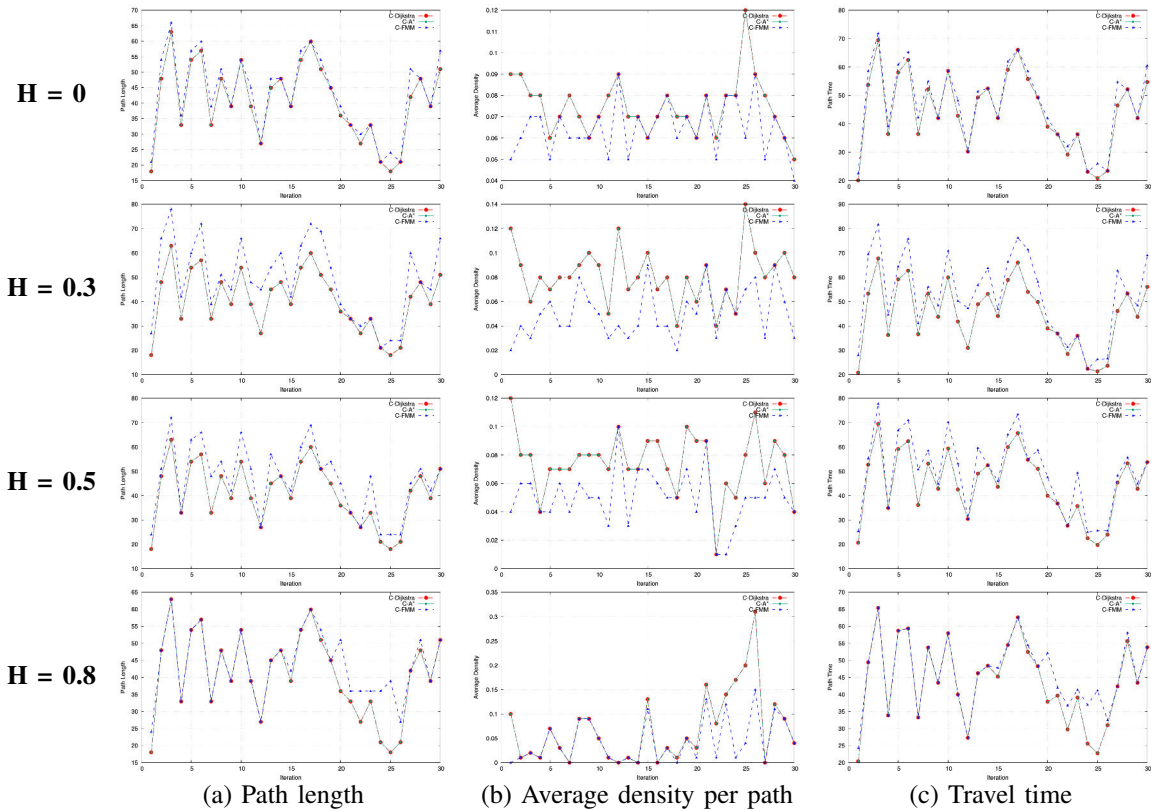
This paper benchmarked Dijkstra, A*, and FMM and their crowd-aware variants for preference-aware indoor navigation on LiDAR-derived maps with accessibility constraints. Using a controlled simulator with synthetic crowds of varying heterogeneity, we quantified trade-offs between path length, crowd exposure, travel time, and computation time.

Overall, crowd-aware planning consistently reduces exposure to congested regions. C-FMM provides the strongest density reduction but often requires longer detours, while C-A* and C-Dijkstra achieve moderate avoidance with minimal change in path length. These results suggest that planner selection should be guided by whether crowd avoidance or travel efficiency is prioritized.

Future work will extend the framework to multi-floor buildings and incorporate real crowd estimation to replace synthetic density fields.

REFERENCES

- [1] N. Abdullah and T. Hua, "Weighted methods of multi-criteria via Dijkstra's algorithm in network graph for less congestion, shorter distance and time travel in road traffic network," *Global Journal of Engineering Science and Researches*, vol. 5, pp. 46–57, 07 2018.
- [2] Y. A. Alaska, A. D. Aldawas, N. A. Algerian, Z. A. Memish, and S. Suner, "The impact of crowd control measures on the occurrence of stampedes during mass gatherings: The hajj experience," *Travel medicine and infectious disease*, vol. 15, pp. 67–70, 2017.
- [3] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, pp. 269–271, 1959.
- [4] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE transactions on Systems Science and Cybernetics*, vol. 4, pp. 100–107, July 1968.
- [5] J. A. Sethian, "A fast marching level set method for monotonically advancing fronts," *proceedings of the National Academy of Sciences*, vol. 93, no. 4, pp. 1591–1595, 1996.
- [6] K.-W. Chiang and Y.-W. Huang, "An intelligent navigator for seamless ins/gps integrated land vehicle navigation applications," *Applied Soft Computing*, vol. 8, no. 1, pp. 722–733, 2008.
- [7] T. Zhang and X. Xu, "A new method of seamless land navigation for gps/ins integrated system," *Measurement*, vol. 45, no. 4, pp. 691–701, 2012.
- [8] P. Melchior, B. Orsoni, O. Lavielle, A. Poty, and A. Oustaloup, "Consideration of obstacle danger level in path planning using a and fast-marching optimisation: comparative study," *Signal processing*, vol. 83, no. 11, pp. 2387–2396, 2003.

Fig. 5: Comparison of base algorithms with heterogeneity level of $H = 0$ Fig. 6: Comparison of crowd-aware algorithms with crowd heterogeneity level of $H = \{0, 0.3, 0.5, 0.8\}$

- [9] T. Liu, H. Li, H. Lu, M. A. Cheema, and L. Shou, "Towards crowd-aware indoor path planning (extended version)," *arXiv preprint arXiv:2104.05480*, 2021.
- [10] L. Liu, S. Zlatanova, and R. Xie, "Pedestrian path choice in crowded indoor environments," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 48, pp. 665–671, 2024.
- [11] P. Cardoso, A. P. Cláudio, and D. Domingos, "An indoor navigation system for reduced mobility users.," in *VISIGRAPP (1: GRAPP)*, pp. 287–298, 2020.
- [12] W. Xu, L. Liu, S. Zlatanova, W. Penard, and Q. Xiong, "A pedestrian tracking algorithm using grid-based indoor model," *Automation in Construction*, vol. 92, pp. 173–187, 2018.
- [13] P. Duszak, B. Siemiatkowska, and R. Wieckowski, "Hexagonal grid-based framework for mobile robot navigation," *Remote Sensing*, vol. 13, no. 21, p. 4216, 2021.
- [14] Polycam, Inc., "Polycam: 3D Scanner & Editor." Mobile Application, 2025. Used for 3D capture via LiDAR sensors on iPhone 16 Pro.
- [15] Polycam, "Why isn't space and floorplan mode not capturing my curved wall?," Polycam Help Center, 2025. [Online]. Available: <https://learn.poly.cam/hc/en-us/articles/30549910889748-Why-Isn-t-Space-and-Floorplan-Mode-Not-Capturing-My-Curved-Wall>. [Accessed: Aug. 28, 2025].
- [16] B. M. Varghese, A. Unnikrishnan, and K. Jacob, "Spatial clustering algorithms-an overview," *Asian Journal of Computer Science and Information Technology*, vol. 3, no. 1, pp. 1–8, 2013.
- [17] J. Montufar, J. Arango, M. Porter, and S. Nakagawa, "Pedestrians' normal walking speed and speed when crossing a street," *Transportation research record*, vol. 2002, no. 1, pp. 90–97, 2007.
- [18] S. E. Sonenblum, S. Sprigle, and R. A. Lopez, "Manual wheelchair use: bouts of mobility in everyday life," *Rehabilitation research and practice*, vol. 2012, no. 1, p. 753165, 2012.
- [19] Employment and Social Development Canada, "Towards an accessible Canada." <https://www.canada.ca/en/employment-social-development/programs/accessible-canada.html>, 2023. Date modified: September 1, 2023. [Online]. Available: <https://www.canada.ca/en/employment-social-development/programs/accessible-canada.html>.
- [20] J. Lee and Y. Kang, "A dynamic algorithm for measuring pedestrian congestion and safety in urban alleyways," *ISPRS International Journal of Geo-Information*, vol. 13, no. 12, p. 434, 2024.
- [21] L. Boodlal, "Accessible sidewalks and street crossings: an informational guide," tech. rep., United States. Federal Highway Administration, 2003.